



PART 1 — LEVEL 1 (BASIC THEORY / BOARD PATTERN)

Electrostatics – 150 Questions

◆ SECTION 1: Electric Charges & Properties (Q1–Q20)

1. Define electric charge.
2. State the SI unit of electric charge.
3. What are the basic properties of electric charge?
4. Explain “quantisation of charge.”
5. Define “conservation of charge.” Give an example.
6. What are positive and negative charges?
7. What is meant by additive property of charges?
8. State the difference between like and unlike charges.
9. Can a charged body attract an uncharged body? Why?
10. What do you mean by charging by conduction?
11. What do you mean by charging by friction?
12. What is meant by charging by induction?
13. Define an isolated charge.
14. What is meant by point charge?
15. Explain free electrons.
16. Why is the charge on an electron always the same?
17. Name the instrument used to detect charge.
18. What is a triboelectric series?
19. Why does a charged balloon stick to a wall?
20. Give two differences between positive and negative charges.

◆ SECTION 2: Conductors, Insulators & Induction (Q21–Q40)

21. What is a conductor? Give two examples.
22. What is an insulator? Give two examples.
23. Define dielectric.
24. State two differences between conductor and insulator.
25. What are semiconductors?
26. Explain the process of charge induction.
27. Why is induction preferred over friction for charging sensitive devices?
28. What happens to charges on a conductor in equilibrium?
29. Why do charges reside on the surface of a conductor?
30. Why is the electric field inside a conductor zero?
31. Define grounding.

32. What is earthing?
33. Explain why metals are good conductors.
34. What is polarization of charge in insulators?
35. Why does a comb attract small pieces of paper after rubbing?
36. What is a hollow conductor?
37. State the shielding effect of a conductor.
38. Why is a lightning conductor used?
39. What is surface charge density?
40. Why does a sharp metallic point leak charge easily?

◆ SECTION 3: Coulomb's Law (Q41–Q60)

41. State Coulomb's law.
42. Write the mathematical form of Coulomb's law.
43. Define the SI unit of force in electrostatics.
44. What is the meaning of the term "electrostatic force"?
45. What is the value of Coulomb's constant?
46. Define linearity of electric forces.
47. On what factors does the electrostatic force depend?
48. Does the electrostatic force follow Newton's third law?
49. What is the nature of electrostatic force between like charges?
50. What is the nature of electrostatic force between unlike charges?
51. Explain the concept of superposition principle.
52. Write the vector form of Coulomb's law.
53. Define medium permittivity.
54. Explain relative permittivity.
55. Why is electrostatic force smaller in water than in air?
56. What is meant by a central force?
57. How is Coulomb's law similar to Newton's law of gravitation?
58. How is Coulomb's law different from gravitational law?
59. What are the limitations of Coulomb's law?
60. What is effective force on a charge placed between two equal charges?

◆ SECTION 4: Electric Field & Field Lines (Q61–Q90)

61. Define electric field.
62. Write the SI unit of electric field.
63. Give the mathematical expression for electric field due to a point charge.
64. What is a test charge?
65. What is meant by electric field intensity?

66. Define electric field lines.
67. State two properties of electric field lines.
68. Why do electric field lines never intersect?
69. What is a radial field?
70. What is a uniform electric field?
71. Draw electric field lines for a positive point charge.
72. Draw electric field lines for a negative point charge.
73. Draw field lines for an electric dipole.
74. What is meant by tangential property of field lines?
75. Define the term "electric flux density."
76. What is the physical significance of electric field?
77. What is neutral point?
78. What is the relation between force and electric field?
79. Define electric field due to an infinitely long charged wire.
80. State the expression for electric field due to a ring of charge on its axis.
81. What is electric field inside a charged spherical shell?
82. What is electric field outside a charged spherical shell?
83. What is an electric monopole?
84. Why does electric field represent direction of force on a positive charge?
85. What is meant by electric field mapping?
86. What is space charge?
87. Can electric field lines be curved?
88. What does closeness of field lines indicate?
89. What does divergence of field lines indicate?
90. What is electric field due to a dipole at axial position?

◆ SECTION 5: Electric Dipole & Dipole Moment (Q91–Q110)

91. What is an electric dipole?
92. Define dipole moment.
93. What is the SI unit of dipole moment?
94. Give two examples of natural dipoles.
95. Write expression for electric field on axial line of dipole.
96. Write expression for electric field on equatorial line of dipole.
97. What is the direction of dipole moment?
98. What is torque on a dipole in uniform electric field?
99. When is torque on dipole maximum?
100. When is torque on dipole zero?
101. What is the physical significance of dipole moment?
102. Explain neutral point in dipole.
103. What is dipole in external electric field?

104. Define polarized molecule.
105. What is nonpolar molecule?
106. What is the behaviour of dipole in non-uniform electric field?
107. What is dipole approximation?
108. What is effective length of dipole?
109. What do you mean by permanent dipole?
110. What is induced dipole?

◆ SECTION 6: Electric Flux (Q111–Q125)

111. Define electric flux.
112. Write the SI unit of electric flux.
113. What is meant by solid angle?
114. What is differential flux area?
115. State the relation between electric field and flux.
116. What is the significance of electric flux?
117. When is electric flux maximum?
118. When is electric flux zero?
119. Define angle of dipole moment with area vector.
120. What is area vector?
121. Write the formula for flux through a plane surface in uniform electric field.
122. What is electric flux through closed surface?
123. What is positive flux?
124. What is negative flux?
125. State any two properties of flux.

◆ SECTION 7: Gauss's Law & Applications (Q126–Q150)

126. State Gauss's law in electrostatics.
127. Write mathematical form of Gauss's law.
128. What is Gaussian surface?
129. What is symmetrical charge distribution?
130. Why is Gauss's law useful?
131. What is electric field for infinite line charge?
132. State application of Gauss's law for infinite plane sheet.
133. State application of Gauss's law for spherical shell.
134. What is electric field inside a uniformly charged sphere?
135. What is electric field outside a uniformly charged sphere?
136. Why Gaussian surface should be symmetric?
137. What is flux through a cylindrical surface?

138. State limitations of Gauss's law.
139. What is meant by non-conducting solid sphere?
140. Plot E vs r graph for a charged spherical shell.
141. Plot E vs r graph for a uniformly charged solid sphere.
142. What is linear charge density?
143. What is surface charge density?
144. What is volume charge density?
145. Define E -field for two infinite sheets.
146. Explain why field inside a hollow conductor is zero using Gauss's law.
147. What is electric field due to charged cylinder?
148. What is flux through a cube containing a charge?
149. What is flux through a cube when charge lies outside?
150. Why does Gauss's law not give direction of electric field?